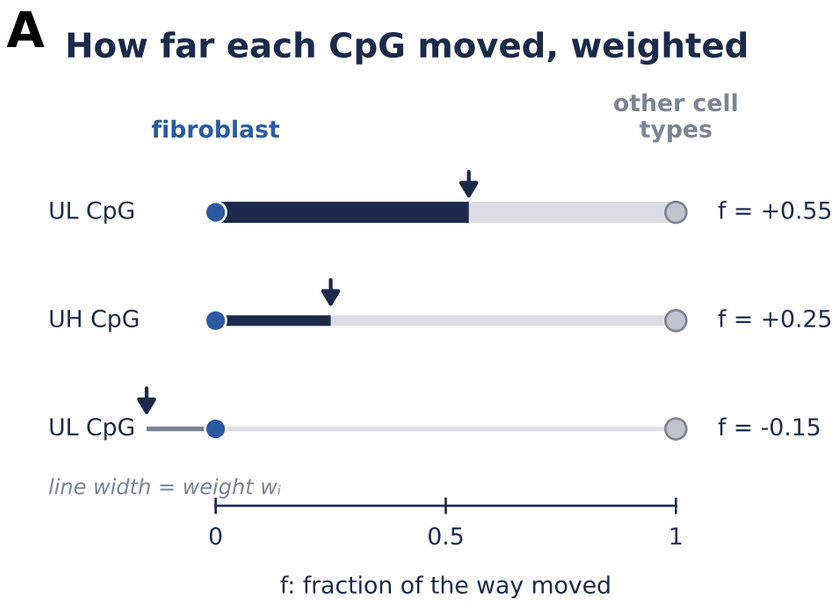
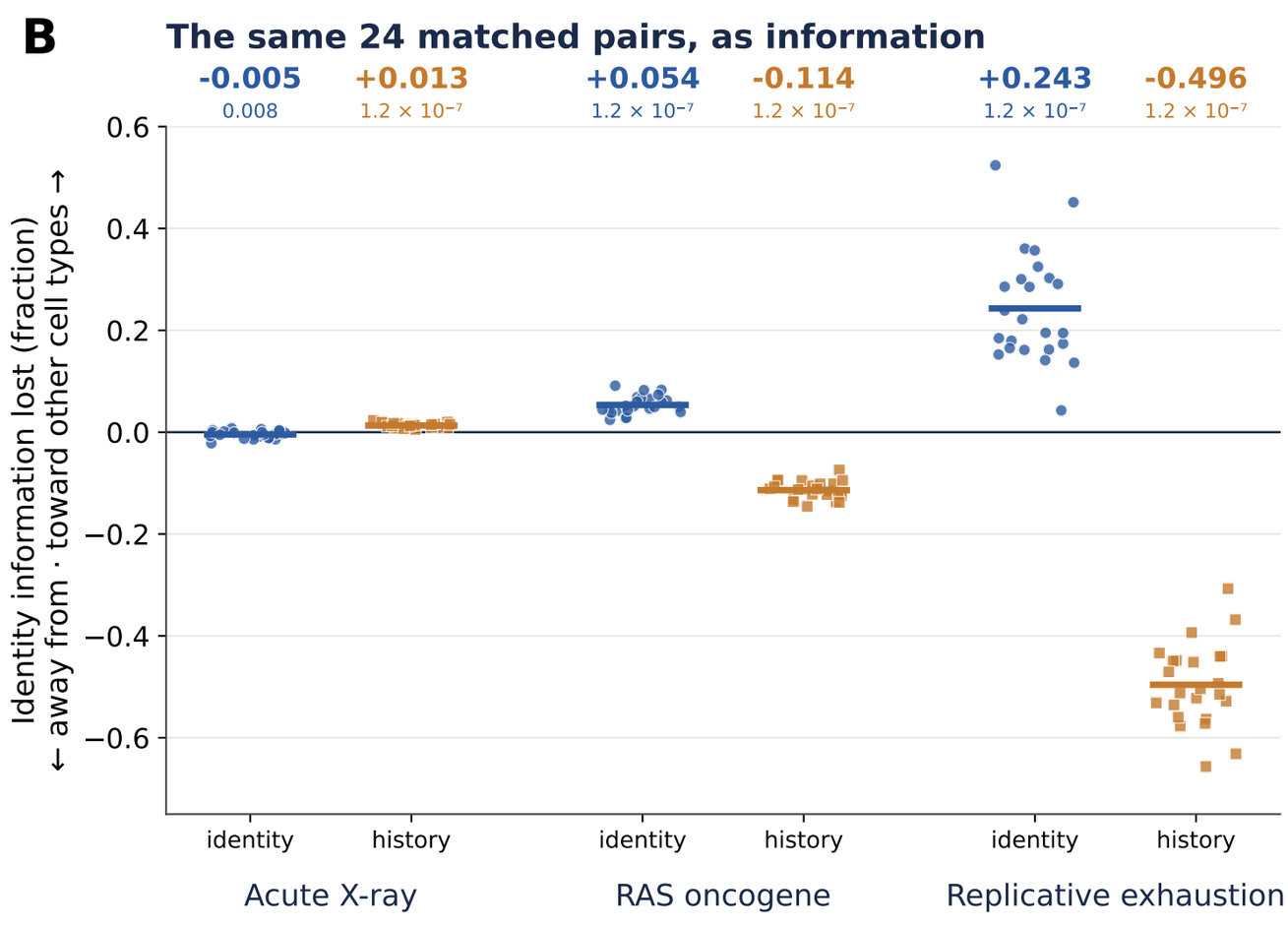
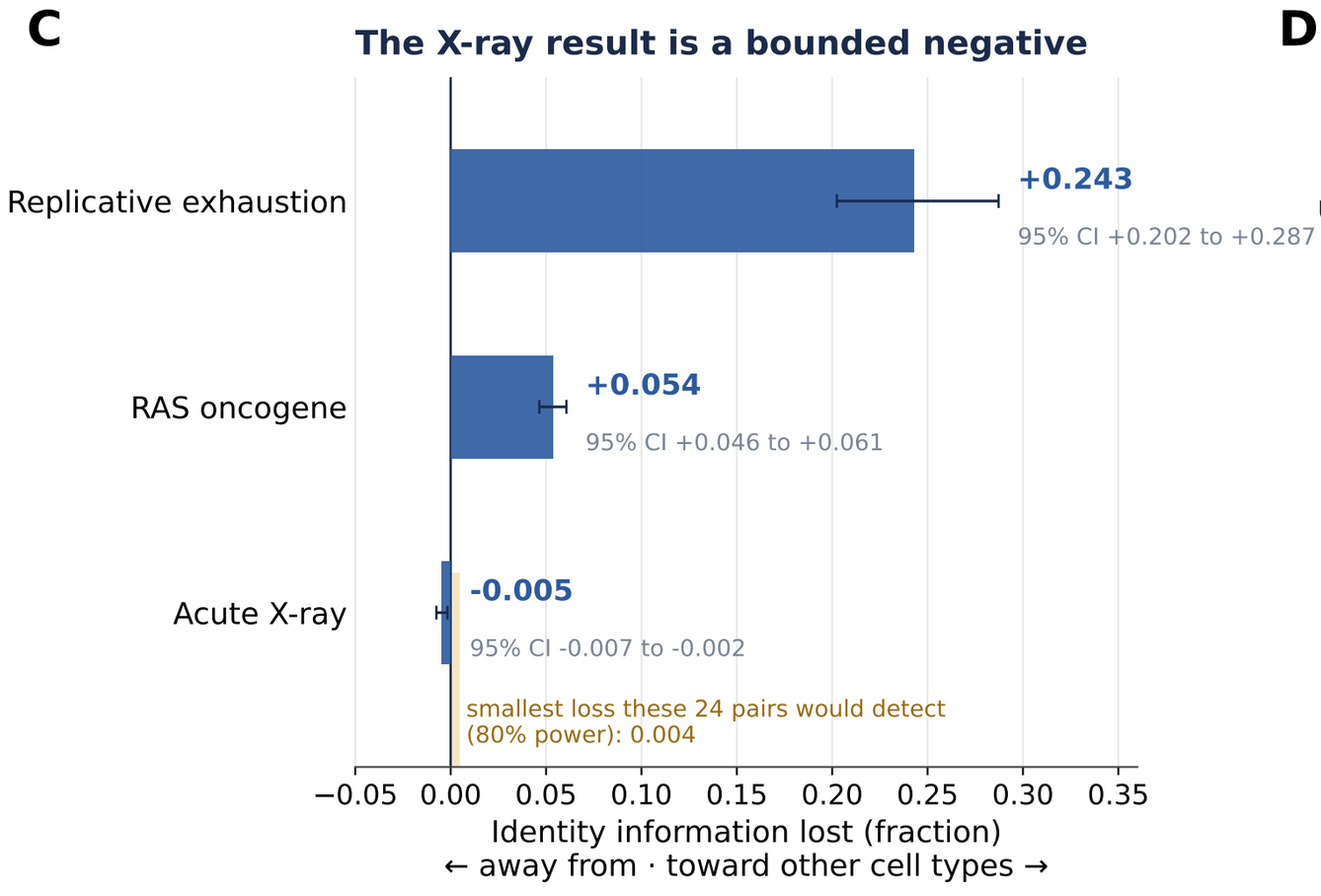
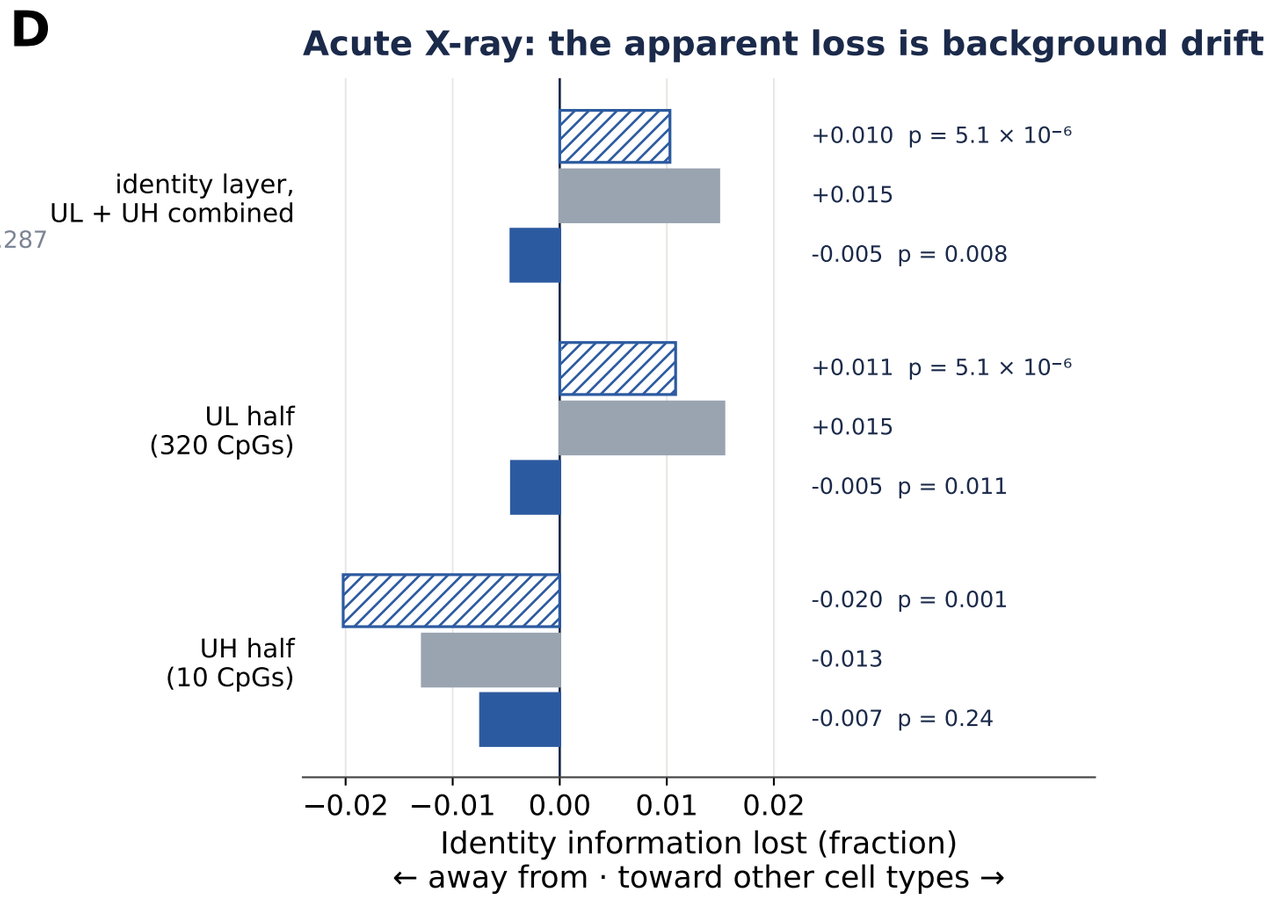

A
How far each CpG moved, weighted
fibroblast
other cell types
UL CpG
f = +0.55
UH CpG
f = +0.25
UL CpG
f = -0.15
line width = weight wᵢ
0
0.5
1
f: fraction of the way moved
information lost = Σ wᵢ fᵢ / Σ wᵢ − background


B
The same 24 matched pairs, as information
-0.005
0.008
+0.013
1.2 × 10⁻⁷
+0.054
1.2 × 10⁻⁷
-0.114
1.2 × 10⁻⁷
+0.243
1.2 × 10⁻⁷
-0.496
1.2 × 10⁻⁷
Identity information lost (fraction)
← away from · toward other cell types →
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
identity
history
identity
history
identity
history
Acute X-ray
RAS oncogene
Replicative exhaustion


C
The X-ray result is a bounded negative
Replicative exhaustion
+0.243
95% CI +0.202 to +0.287
RAS oncogene
+0.054
95% CI +0.046 to +0.061
Acute X-ray
-0.005
95% CI -0.007 to -0.002
smallest loss these 24 pairs would detect
(80% power): 0.004
−0.05
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Identity information lost (fraction)
← away from · toward other cell types →


D
Acute X-ray: the apparent loss is background drift
identity layer,
UL + UH combined
+0.010 p = 5.1 × 10⁻⁶
+0.015
-0.005 p = 0.008
UL half
(320 CpGs)
+0.011 p = 5.1 × 10⁻⁶
+0.015
-0.005 p = 0.011
UH half
(10 CpGs)
-0.020 p = 0.001
-0.013
-0.007 p = 0.24
−0.02
−0.01
0.00
0.01
0.02
Identity information lost (fraction)
← away from · toward other cell types →
panel CpGs' own shift
background term
net = panel − background